Revision log for "On quadratic residue codes and hyperelliptic curves"

Corrected and expanded version, Draft 1

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November 11, 2026

Scope

Original: `original/qrcodes-hyperell-dmtcs2.tex`, the version submitted to DMTCS (received 20 October 2006), sole author D. Joyner.
Revised: `qrcodes-hyperelliptic.tex`, D. Joyner and Claude. Current draft: Draft 1 (November 11, 2026).

Numbering. Sections 1–6 were written before Draft 1 and use the numbering of that time. Draft 1 moved the literature review to Section 2, so most numbers have changed. Section 7.5 gives a concordance, and the map of results in Section 11 uses the Draft 1 numbers.

The published 2008 DMTCS version (doi:10.46298/dmtcs.429, arXiv:math/0609562v2) is a *different* revision of the same submission. This log compares only against the 2006 file. In the original, numbering ran with one counter per environment; in the revision, numbering is by section (for example, Theorem 4.2). The table in Section 11 maps old results to new ones.

Every numerical claim in either version was checked with the Python code in `code/`, using exhaustive enumeration of C_{NQ} for $p \leq 23$.

1 Format and front matter

1. Document class changed from `dmtcs` to `article`, 12pt, letter paper, Times (`mathptmx`), compiled with `pdflatex`. The RCS macros, `latin1` input encoding, `subfigure` and `natbib` were removed.

2. Author block: the affiliation "Mathematics Dept., USNA, Annapolis, MD" was replaced by `wdjoyner@gmail.com`. Claude (Anthropic) was added as second author, with a footnote and an acknowledgment paragraph saying what it contributed.

3. The abstract was rewritten to state the corrected results and constants. Keywords were updated, and the MSC codes were updated to MSC 2020.

4. A table of contents was added.

5. The title-page date "Original submission: 20 October 2006" was replaced by "November 2026". The paragraph describing the relation to the 2006 version (at first Section 1.5, *About this revision*) was moved to an unnumbered section just before the Acknowledgments. Apart from that paragraph, the revised paper no longer refers to the 2006 version; all such comparisons are in this log.

6. The Acknowledgments now point to this revision log.

7. Layout: the weight-distribution table (Table 1) and Remark 4.7 were reset so that the PDF has no overfull lines.

8. The material after `\end{document}` (an unused remark with an unproved lower bound on $|S_3|$) was deleted.

2 Mathematical corrections

2.1 Errors corrected

1. **Lemma 1 (old), parity claims.** We have $r^*_{S_1} r^*_{S_2} = r_{S_1} r_{S_2} + (|S_1| + |S_2| + 1)J$, and $\mathrm{wt}(r_{S_1} r_{S_2}) \equiv |S_1||S_2|$. So in the cases (even, even) and (odd, odd) the element $(r_{S_1} r_{S_2})^*$ has odd and even weight respectively, the opposite of what was stated. The lemma was replaced by the short list of facts in §2.1.

2. **"$*$ is an algebra involution on R_{even}."** This is false: $*$ is not additive, since $(a+b)^* = a+b+J$, and it maps even-weight elements to odd-weight ones. Also, $R_{even} = (1+x)R$ contains no 1. The claim was deleted.

3. **Remark 1 (old).** "C_{NQ} contains a QR code" was corrected to "$C_{NQ} \subseteq \mathrm{QR} \oplus \mathrm{QR}'$" (Remark 3.12).

4. **Lemma 2 (old)** relied on self-duality, which was stated only as a conjecture (Conjecture 3, old). Self-duality is now proved (Theorem 3.4), so the lemma is justified. The bound is now attributed correctly to Rains [Ra98], not Sloane–Mallows.

5. **Corollary 1 (old),** $\max \#X_S > \frac{5}{3}p - 4$**.** The proof assumed that a minimum-weight codeword comes from an even-size S. For $p = 11$ it does not: the 77 weight-6 words all come from odd S. The result was restated and reproved as Corollary 4.5, $M_p \geq \frac{5}{3}p - 6$ for $p \equiv 3 \pmod 4$ and $p \geq 11$, using the new twist identity.

6. **Remark after Proposition 1 (old).** The text gave both "$\sqrt{p} + 1$" and "$\sqrt{p} + 2$", and applied part (c) to the wrong parity. The corrected statement is Remark 3.12. That remark also notes that the bound follows from Weil's bound and the trivial count. (The referee report said this bound was stronger than Weil; that was wrong and is corrected here.)

7. **Theorem (old 2), proof.** The entropy function had a spurious leading δ. The weight formula now cites Proposition 1, not Lemma 3.

8. **Theorem (old 3), proof.** The proof claimed to contradict "$B(1.62, p)$ for all sufficiently large p". The correct logic is that $B(1.62, p)$ can hold for only finitely many p. Rewritten as Theorem 4.3 with the sharp constant $c^* = 1.62584\ldots$.

9. **Theorem 4 (old, LQR parameters).** For $p \equiv 3 \pmod 4$, the claimed lower bound $d \geq 4p - 2\max \#X_S$ was not proved. The injectivity argument was incorrect: it asserted $r_Q r_{T^c} = 0$, and it concluded "$S_1 = \emptyset$, $S_2 = \mathbb{F}_p$", contradicting the injectivity it claimed. For $p \equiv 1 \pmod 4$, d_p should be $4p + 4 - 2M$. The statement "this code is non-linear" is false for $p \equiv 1 \pmod 4$.

10. **Remark after Theorem 4 (old).** The remark doubted that the minimum distance could be below $2p$. In fact it always is: the minimum distance is $2d(C_{NQ})$, for example $8 < 14$ at $p = 7$.

11. **Lemma 7 (old).** For $p \equiv 1 \pmod 4$ and odd S, the sum should be over f_{S^c}.

12. **Example (old, LQR, $p = 11$).** The two sets $S = \{1, 2, 3, 4\}$ and $\{1, 2, 3\}$ were interchanged. The section was removed (see Section 3).

13. **Section 3 (old), genus.** The formula $\gamma = n + k + 1 - d$ was corrected to $\gamma = n + 1 - k - d$.

14. **Example (old, $p = 13$ zeta).** The text said "only 8 of the 12 zeros". $P(T)$ has degree 16, and exactly 12 of its 16 zeros lie on $|T| = 1/\sqrt{2}$. The coefficients were correct.

15. **Optimality of the $[26, 13, 6]$ code.** The claim rested on general linear-code tables, which, as we understand them, allow $d = 7$ for $[26, 13]$. It was restated as optimality among even codes, with an explicit caveat that this has not been verified independently.

16. **Voloch's Theorem (old 6).** The constant window $-\frac{1}{2} \leq a \leq \frac{5}{2}$ holds only for $\ell = 2$. It was replaced by $O(1) \in [-2, 3]$, and "$\ell > 2$" versus "$\ell \geq 2$" was made consistent. The introduction's claim that $\#X_Q > 1.5p$ for $p \equiv 3 \pmod 8$ was corrected to $\#X_Q = 1.5p - \frac{1}{2}$.

17. **Conjecture 3 (old).** "$C \cap C^\perp = \emptyset$" was corrected to $\{0\}$, and "direct product" to "direct sum".

2.2 Former conjectures now proved

1. Conjecture 3 (old): self-dual for $p \equiv 3 \pmod 4$, LCD for $p \equiv 1 \pmod 4$. Now Theorem 3.4, with a three-line proof via $\langle u, v \rangle = \mathrm{ct}(uv^-)$.

2. Conjecture 4 (old): C' is formally self-dual with distribution $A + A^*$. Now Theorem 6.2: $C'^\perp$ is the half-swap of C', so C' is isodual. The distribution statement is immediate.

3. Section 3 (old) asked whether C_{NQ} always satisfies RH for $p \equiv 3 \pmod 4$. Answered negatively: RH fails at $p = 23$ (30 of 32 zeros on the circle). This agrees with Boston–Hao [BH17].

2.3 New results

1. Twist identity (Lemma 2.4): $\#X_T + \#X_{T'} = 2p+2$ when $\chi(f_T(a)) = -1$, and its consequence $M_p + m_p \geq 2p + 2$ (Corollary 2.6).

2. Even subcode C_{ev} (Proposition 3.10). It has dimension $p-1$ for every p, weights $2p+2-\#X_T$, and $d(C_{\mathrm{ev}}) = 2p + 2 - M_p$. For $p \equiv 3 \pmod 4$ it is the doubly-even subcode, so $\#X_T \equiv 0 \pmod 4$ for even T.

3. Theorem 4.1 (Goppa): the congruence condition $p \equiv 1 \pmod 4$ was removed, and the threshold sharpened to $c < 1.7799$. This supersedes the LQR result (old Theorem 5, $c = 1.57$).

4. Theorem 4.3 (MRRW): valid for all p, with $c^* = 1.62584$.

5. Theorem 5.1(3): the case $p \equiv 5 \pmod 8$, $S = Q \cup \{0\}$, which gives $\#X_S = 1.5p + \frac{1}{2}$. Every prime $p \geq 11$ now has an explicit S with $\#X_S \geq 1.5p - \frac{1}{2}$.

6. New data: weight distributions for $p = 17, 19, 23$ (Table 1), the RH status of seven codes (Table 2), and M_p, m_p for $p \leq 23$ (Table 4; Table 3 before the addition of 10 November 2026).

3 Material removed

1. Section 4 (old), *Long quadratic residue codes*: Lemmas 5–7, Theorem 4, Theorem 5, and the LQR example. They were replaced by Remark 4.7, which explains that the LQR code is a doubled code or a coset of one, and so has the same relative distance as C_{NQ} at half the rate. The even subcode gives the stronger result.

2. The "Cyclotomic arithmetic" section was condensed to the facts actually used. The sets $H(S_1, S_2, a)$ and identity (2) (old) were removed.

3. The quoted lemma "Bazzi–Mitter Prop. 3.3" (old Lemma 4) was absorbed into Proposition 3.6.

4 Additions for readability

1. Eleven worked examples (Examples 2.1, 2.3, 2.5, 3.3, 3.7, 3.8, 3.11, 6.1, 6.3 and two in Sections 4 and 5). They cover arithmetic in R, point counting via Legendre sums, the twist identity, dimension, the weight formula, the even subcode, Voloch's construction for $p = 13$, and zeta polynomials.

2. Proofs were rewritten in full, including the dimension count (Proposition 3.2) and Lemma 2.2 (Artin–Hasse–Weil) with the trivial bounds (3).

3. Section 8 (Section 7 before 10 November 2026), *Open questions and their difficulty*: a random-code heuristic ($M_p \approx 1.78p$) and an assessment of each open question.

4. Section 9 (formerly Section 8), *Literature*, in two parts: before 2006 and 2006 and after.

5. Appendix A, *Glossary*: 54 entries with a short introduction. It is placed after the Acknowledgments and before the bibliography.

6. An introductory paragraph was added at the start of every section and subsection, saying what that part contains and what to expect.

7. In the list of results, "answers Question 1.4 for" now reads "answers Question 1.4 affirmatively for".

8. Typos fixed, among others: McEliese → McEliece, Welsh → Welch, Tsafsman → Tsfasman, Dordrechet → Dordrecht, "This rest of", "the the", "if and only if and only if", "consquence", "we if"; the cross-reference "Theorem \ref{cor:main}"; and the inconsistent "$p > 5$". The second supplementary law is no longer called "quadratic reciprocity".

5 Bibliography

1. Updated: Bazzi–Mitter (IEEE IT 2006), Jiang–Vardy (IEEE IT 2004), Joyner (DMTCS 2008).

2. Added: Alahmadi–Özdemir–Solé 2018, Boston–Hao 2017, Calderbank 1983, Chen–Peterson–Weldon 1969, Chinen 2017/18, Grassl tables, Joyner–Kim 2011, Karlin 1969, Kasami 1974, Mallows–Sloane 1973, Massey 1992, MRRW 1977, Rains 1998, Shi–Xu–Solé 2021.

3. Removed as uncited: SAGE 2.10, Wage 2006, Chinen's "An abundance of invariant polynomials". The "Wage" remark on $|S| = 2, 3$ was dropped.

4. Bibliographic details marked from memory in the referee report should be checked against the publishers' pages before submission.

6 Additions of 10 November 2026

These changes came out of a further discussion of the Bazzi–Mitter conjecture. They add a section and some data; nothing earlier was withdrawn.

1. New Section 7, *The split by p modulo 8*. It is placed before *Open questions*, so the later sections are renumbered: Open questions is now Section 8, Literature Section 9.

   > Proposition 7.1: for $p \equiv \pm 1 \pmod 8$, C_{ev} is the direct sum of two even-like binary QR codes, so $M_p = 2p + 2 - d(\widehat{Q}_p)$ exactly.
   >
   > Proposition 7.2: for $p \equiv \pm 3 \pmod 8$, C_{ev} is $\mathbb{F}_4$-linear and is the binary image of an even-like quaternary QR code, with $d_4 \le d(C_{\mathrm{ev}}) \le \frac{4}{3} d_4$. The bordered analog is due to Karlin, Bhargava and Tavares (1978).
   >
   > Remark 7.3: the pure and bordered codes can differ ($p = 13$: $d(C_{\mathrm{ev}}) = 8$, $d(B_{13}) = 6$).

Theorem 7.4: Conjecture 1.3 implies linear minimum distance for binary or quaternary QR codes along an infinite set of primes.

Conjecture 7.5 (new, refined): there is $c < 2$ with $M_p \le cp$ for infinitely many $p \equiv \pm 3 \pmod 8$. Remark 7.6 records that it is equivalent to linear distance for quaternary QR codes, that $c < 1.7799$ would refute Goppa's conjecture, and a stronger form for safe primes $p \equiv 3 \pmod 8$.

Table 3 (new): exact $d(C_{\mathrm{ev}})$ and M_p for all $p \equiv \pm 3 \pmod 8$ with $p \le 83$, bounds for $p = 107$, and exact M_p for $p \equiv \pm 1 \pmod 8$, $p < 200$, from the extended-QR distances of Tomlinson et al. (2017, Table 9.4).

Remark 7.7: the Legendre-symbol form $\#X_T = |T| + 2 + 2N_+(T)$, and the outline of a possible proof that separates structured from unstructured sets.

2. Abstract: a closing sentence on the new section. List of results: item 8. After Conjecture 1.3: a pointer to Section 7.

3. Open questions: the heuristic paragraph now says that the $1.78p$ heuristic fits only $p \equiv \pm 3 \pmod 8$. The Bazzi–Mitter paragraph cites Theorem 7.4. New paragraphs on Conjecture 7.5 and on analytic hypotheses (GRH, Hooley).

4. Literature: Karlin–Bhargava–Tavares 1978 and Gaborit 2002 (before 2006); Tomlinson–Tjhai–Ambroze–Ahmed–Jibril 2017 (after 2006). The About-this-revision paragraph and the Acknowledgements mention the new section.

5. Glossary: new entries *Binary image*, *Bordered double-circulant code*, *Even-like code*, $N_{\pm}(T)$; the entry *QR code* now covers quaternary QR codes.

6. Bibliography: added Gaborit 2002, Hooley 1967, Karlin–Bhargava–Tavares 1978, Tomlinson et al. 2017.

7. Code: `code/check_mod8.py` checks Propositions 7.1 and 7.2 for small p; `code/dmin_cev.c` computes $d(C_{\mathrm{ev}})$ exactly for $p \equiv \pm 3 \pmod 8$ (Table 3).

7 Draft 1: revision for the style guide (November 11, 2026)

7.1 Scope

Claude brought the paper into line with D. Joyner's style guide (`joyner-math-paper-style-guide.md`, Draft 2 of November 11, 2026), following his instructions of the same day. Those instructions forbade mathematical changes. No statement, proof, hypothesis or status label was altered. All new mathematical text was confined to what the guide requires, and it is listed in Section 7.4 so that it can be refereed. Semicolons in existing text were left in place. Rule 3.8 (no semicolons) was applied to new text only. The guide has no paper-specific rule (its Section 9) for this paper.

Before the changes, the paper compiled with no errors, no undefined references and no overfull boxes. There were 7 hyperref warnings, all caused by math in the title of the subsection on the extended code. The typewriter font was a bitmap font. After the changes, the paper compiles with no errors and no warnings, and every font is Type 1.

7.2 Defaults used for the rules marked [DECIDE]

1. Document class: the existing class, `article`, was kept.
2. Times package: `newtxtext` and `newtxmath`.
3. Glossary placement: notation near the beginning (a table at the end of Section 3), and the glossary at the end, just before the Acknowledgments.
4. AI systems on the author line: the instructions did not mention the author line, so it was left as it was (D. Joyner and Claude). The Acknowledgments also credit Claude and point to this log.

7.3 Changes

Typesetting.

1. `mathptmx` was replaced by `newtxtext,newtxmath`. The package `amssymb` was dropped because it conflicts with `newtxmath`, which supplies the same symbols. The same change was made in this log.
2. Equations are numbered within sections (`\numberwithin`), as rule 1.6 requires. So (1)–(4) became (3.1), (3.2), (3.3) and (4.1).
3. `\mathrm{circ}` became `\operatorname{circ}` (rule 1.8).
4. The title of the subsection on the extended code uses `\texorpdfstring`, which removes the 7 hyperref warnings.
5. The hand-typed "Item 6" in Section 1 became a `\ref` to a new label `item:voloch` (rule 1.5).

Title page, abstract and acknowledgments.

1. The subtitle "Revised and updated" became "Corrected and expanded version. Draft 1" (rule 6.2). The title itself is unchanged.
2. The date line "November 2026" became "Original draft: 2006. Corrected and expanded draft: 2026." (rule 6.2).
3. A `\thanks` note on the title identifies the 2006 DMTCS submission and the published revision [Jo08], and gives the date of Draft 1 (rule 6.3).
4. The abstract's first sentence, "This AI-aided paper revises and updates the first author's earlier paper [Jo08]", was replaced by an AI-disclosure sentence that refers to the Acknowledgments (rule 5.2). The old sentence was also inaccurate: the revision is of the 2006 submission, not of [Jo08] (see Scope at the top of this log). The rest of the abstract is unchanged.
5. "Acknowledgements" became "Acknowledgments" in the heading, in the footnote on Claude, and in this log (rule 1.4).

6. The Acknowledgments now say that Claude acted as referee, reviser and co-author (rule 5.3). They describe Claude's role in Draft 1, state that AI review is not an independent certification of the results (rule 5.4), and say that further details on the AI usage are in this log (rule 5.5). They give the commit hash of the scripts (rule 8.4).

Citations. Every `\bibitem` now prints its key as its label (for example [Jo08] rather than [17]), and the bibliography is sorted by label (rule 1.7). Five keys were renamed, and every `\cite` was updated. The new keys were chosen by D. Joyner.

Old key	New key	Note
`Gr`	`Gr26`	"accessed 2026" added to the entry
`Vo`	`Vo04`	entry completed, see below
`TT17`	`TTAAJ17`	initials of all five authors
`Coy06`	`Co06`	two-letter initial, as for other single authors
`Ch18`	`Ch17`	matches the year of the arXiv entry cited

The entry `Vo04` was a bare "preprint". It now reads: J. F. Voloch, Asymptotics of the minimal distance of quadratic residue codes, in *Finite Fields: Theory and Applications*, Oberwolfach Report No. 54/2004, 2946–2947. These details come from the copy of that report that D. Joyner pointed to (`https://ems.press/content/serial-article-files/45972`). The entry `Ch17` still cites arXiv:1709.03389 (2017). That paper was later published in *Appl. Algebra Engrg. Comm. Comput.* 30 (2019), 275–284, and the entry could be updated to the journal version.

Structure (Section 2 of the guide).

1. The literature review (old Section 9) moved to Section 2, just after the Introduction. Its text is unchanged. It was already split at 2006, the year of the original manuscript.

2. The old Sections 2–8 became Sections 3–9.

3. Section 3 (*Notation and background*) ends with a new table of notation (Subsection 3.4).

4. The old Section 8, *Open questions and their difficulty*, became Section 9, *Summary of results and open questions*. It begins with a new orienting paragraph and a new Subsection 9.1, *Summary of results*. The old text forms Subsection 9.2, under the old title. Its first sentence now says "This subsection" instead of "This section".

5. The glossary (Appendix A) now comes before the Acknowledgments, not after them.

6. The unnumbered section *About this revision* was removed from the paper (rule 6.4). Its content is now in the `\thanks` note on the title and in this log. Its text was:

> This is a revision of the version submitted to *Discrete Mathematics and Theoretical Computer Science* on 20 October 2006. A different revision was published as [Jo08] (see also arXiv:math/0609562). The present text corrects several errors in the 2006 version, replaces some arguments, proves two statements that were stated as conjectures, and adds examples, a literature survey and a glossary. Section 7, with the refined Conjecture 7.5, was added on 10 November 2026. A separate revision log lists every change.

(In Draft 1 numbering, Section 7 and Conjecture 7.5 are Section 8 and Conjecture 8.7.)

7.4 New text to be refereed

Rules 3.2–3.5 required the following additions. All numbers are Draft 1 numbers. Each numerical claim in N1–N4 is checked by `code/check_examples_draft1.py` (commit `c6e9125`). No new result is claimed. The examples illustrate results already in the paper.

N1. **Example 1.3** (`ex:Bp7`), after Definition 1.2 (rules 3.3 and 3.4). For $p = 7$ it computes $\#X_N = 12$ by hand. An exhaustive search shows that 12 is the largest $\#X_S$ over all 127 non-empty S, attained also by the even set $\{0, 1, 2, 4\}$. So $B(c, 7)$ and $B_{\mathrm{ev}}(c, 7)$ hold exactly when $c \geq 12/7$.

N2. **Example 8.2** (`ex:pm1p7`), after Proposition 8.1 (rules 3.3 and 3.4). For $p = 7$, $r_Q R_{\mathrm{ev}}$ is 0 and the seven cyclic shifts of $x^2 + x^4 + x^5 + x^6$, a $[7, 3, 4]$ code. So $d(C_{\mathrm{ev}}) = 4$ and $M_7 = 12$, and $\widehat{Q}_7$ is the $[8, 4, 4]$ extended Hamming code.

N3. **Example 8.4** (`ex:pm3p5`), after Proposition 8.3 (rules 3.3 and 3.4). For $p = 5$, every nonzero word of C_{ev} has $\mathbb{F}_4$-weight 4 and binary weight 4 or 6. So $d_4 = 4$, $d(C_{\mathrm{ev}}) = 4$ and $M_5 = 8$.

N4. **Example 9.1** (`ex:heuristic`), in Subsection 9.2 (rule 3.3). It evaluates the heuristic constant $2(1 - H^{-1}(1/2)) = 1.7799\ldots$ at $p = 11$ and $p = 23$ (predictions 19.6 and 40.9) against $M_{11} = 16$ and $M_{23} = 40$.

N5. **Orienting paragraphs** at the start of Subsections 8.1, 8.2 and 8.3 (rule 3.2). They summarize what follows. The one for 8.2 says that the binary minimum distance of C_{ev} is determined up to a factor of at most $4/3$, which restates Proposition 8.3.

N6. **Summary of results** (Subsection 9.1) and the opening paragraph of Section 9 (Section 2.7 of the guide). Seven items restate the main results with references. Each states the status of its result as the body does. Item 4 marks Theorem 5.1 as conditional, item 6 marks the zeta results as computational, and item 7 keeps Conjecture 8.7 a conjecture.

N7. **Table of notation** (Subsection 3.4), with an orienting paragraph (Section 2.5 of the guide). It has no caption, so the table numbers are unchanged.

N8. **Definitions at first use** (rule 3.5) in the list of results in Subsection 1.3. Item 1 defines "isodual", item 3 defines H^{-1}, and item 8 defines C_{ev}, M_p and $\widehat{Q}_p$, which were used there before their definitions.

N9. **Front matter**: the `\thanks` note, the first sentence of the abstract, and the new Acknowledgments paragraphs described above.

7.5 Concordance of numbers

The left column gives the numbering just before Draft 1, and the middle column gives the Draft 1 numbering. Every `\label` was kept. The table was generated from the two source files and checked

against the label numbers that LaTeX wrote to the `.aux` files. Sections map as follows: old 1 to new 1, old 2–8 to new 3–9, and old 9 (Literature) to new 2. Table numbers did not change.

Before Draft 1	Draft 1	Label
Conjecture 1.1	Conjecture 1.1	`conj:goppa`
Definition 1.2	Definition 1.2	`def:B`
Conjecture 1.3	Conjecture 1.4	`conj:BM`
Question 1.4	Question 1.5	`q:1`
Question 1.5	Question 1.6	`q:2`
(1)	(3.1)	`eq:minus`
(2)	(3.2)	`eq:wtprod`
Example 2.1	Example 3.1	`ex:p7arith`
Lemma 2.2	Lemma 3.2	`lem:AHW`
(3)	(3.3)	`eq:trivial`
Example 2.3	Example 3.3	`ex:p7points`
Lemma 2.4	Lemma 3.4	`lem:twist`
Example 2.5	Example 3.5	`ex:twist`
Corollary 2.6	Corollary 3.6	`cor:Mm`
Definition 3.1	Definition 4.1	`def:QQR`
Proposition 3.2	Proposition 4.2	`prop:dim`
Example 3.3	Example 4.3	`ex:dim`
Theorem 3.4	Theorem 4.4	`thm:duality`
(4)	(4.1)	`eq:ip`
Remark 3.5	Remark 4.5	(no label)
Proposition 3.6	Proposition 4.6	`prop:main`
Example 3.7	Example 4.7	`ex:p7weights`
Example 3.8	Example 4.8	`ex:p11`
Definition 3.9	Definition 4.9	(no label)
Proposition 3.10	Proposition 4.10	`prop:Cev`
Example 3.11	Example 4.11	`ex:Cev11`
Table 1	Table 1	`tab:wd`
Remark 3.12	Remark 4.12	`rem:sqrt`
Theorem 4.1	Theorem 5.1	`thm:goppa`
Remark 4.2	Remark 5.2	(no label)
Theorem 4.3	Theorem 5.3	`thm:mrrw`
Remark 4.4	Remark 5.4	(no label)
Corollary 4.5	Corollary 5.5	`cor:53`
Example 4.6	Example 5.6	(no label)
Remark 4.7	Remark 5.7	`rem:LQR`
Theorem 5.1	Theorem 6.1	`thm:voloch`
Example 5.2	Example 6.2	(no label)
Theorem 5.3	Theorem 6.3	`thm:chebotarev`
Corollary 5.4	Corollary 6.4	(no label)
Example 6.1	Example 7.1	`ex:zeta7`

Before Draft 1	Draft 1	Label
Theorem 6.2	Theorem 7.2	`thm:isodual`
Example 6.3	Example 7.3	`ex:zeta13`
Table 2	Table 2	`tab:RH`
Proposition 7.1	Proposition 8.1	`prop:pm1`
Proposition 7.2	Proposition 8.3	`prop:pm3`
Remark 7.3	Remark 8.5	(no label)
Theorem 7.4	Theorem 8.6	`thm:mod8`
Conjecture 7.5	Conjecture 8.7	`conj:refined`
Remark 7.6	Remark 8.8	`rem:refined`
Table 3	Table 3	`tab:mod8`
Remark 7.7	Remark 8.9	`rem:legendre`
Table 4	Table 4	`tab:Mp`
—	Example 1.3	`ex:Bp7` (new)
—	Example 8.2	`ex:pm1p7` (new)
—	Example 8.4	`ex:pm3p5` (new)
—	Example 9.1	`ex:heuristic` (new)

7.6 Rules not applied, and why

1. Rule 1.6, second sentence (do not number subsections that do not need a number). All subsections are listed in the table of contents, and this log cites them by number, so they stay numbered.

2. Rule 3.3 in Section 2 (Literature). A worked example there would add nothing.

3. Rule 3.8 in existing text, by instruction. Existing semicolons were kept.

4. Rules 3.6 and 3.7 (hypotheses before theorems, no skipped steps). Applying them would change statements or proofs, which this draft did not do. No violation was looked for.

5. Rules 4.3 and 4.4. No citation was checked against its source, and no computation in the body was rerun, except as described in the next subsection.

6. Rule 6.6 ("Remark 2026.n" for anything not fully verified). Applying it would add status labels, which this draft did not change. One candidate is the optimality of the $[26, 13, 6]$ code after Example 7.3, which the text already qualifies in prose.

7. Rule 1.5 for citations typed by hand. "Gleason–Prange" (proof of Proposition 8.1), "Prange (1957–1959)" and "Gilbert (1952) and Varshamov (1957)" (Section 2) have no `\cite`. Fixing them needs new bibliography entries or precise theorem numbers that were not checked.

8. Rules 8.5 and 8.6 concern the repository, not the paper. The repository has no license file. It is public and contains a corrected version of a manuscript whose different revision was published as [Jo08]. Neither point was acted on.

7.7 Not verified

1. The Voloch abstract (Vo04) was read only through a web tool's summary, not in a printed copy. That summary says the abstract shows that some subfamilies of binary QR codes are asymptotically bad. Section 2 describes that paper only as studying "the asymptotics of QR code minimum distances", and Section 8 says that whether binary QR codes are asymptotically good is open. These statements look compatible, but a referee should compare them with the abstract.
2. The new notation table and summary were checked against the body by reading, not by an independent referee.
3. The bibliographic details of entries other than Vo04 were not rechecked in Draft 1.

8 AI usage

All AI work on this revision was done by Claude (Anthropic), recorded as Claude Opus 5.5 in the paper.

1. *Referee.* Claude refereed the 2006 manuscript. Its findings are the corrections in Section 1 onwards.
2. *Co-author and reviser.* Claude proposed and wrote the proofs and computations listed in the Acknowledgments of the paper, including the 10 November 2026 additions (Section 6). D. Joyner reviewed them.
3. *Programmer.* Claude wrote the scripts in code/, which check every numerical claim.
4. *Style reviser (Draft 1).* Claude made the changes in Section 7 and wrote the new text N1–N9, which has not yet been refereed.

AI review is not an independent certification of the results.

9 Sources not consulted

Rule 4.3 asks that every citation be checked against its source. In Draft 1 no cited source was consulted, apart from the Oberwolfach report for Vo04 (through a web tool) and the publication record of Ch17. The earlier referee report marked some bibliographic details as written from memory (see Section 5, item 4). The optimality claim for the $[26, 13, 6]$ code rests on the tables [Gr26] as the authors understand them, without independent verification.

10 Original and new results

Rule 6.5 asks that original results be separated from results first established in the revision. Section 2 does this. Its subsections list the errors corrected in original results, the former conjectures now proved, and the new results. The map of results in Section 11 gives the correspondence. Draft 1 adds no new results.

11 Map of results

2006 version	Draft 1
Lemma 1 (products)	§3.1 facts (corrected)
Lemma 2 (mindist bound)	used in Corollary 5.5 (Rains)
Lemma 3 (Artin–Hasse–Weil)	Lemma 3.2
Lemma 4 (Bazzi–Mitter)	absorbed into Proposition 4.6
Proposition 1	Proposition 4.6
Corollary 1 ($\frac{5}{3}p - 4$)	Corollary 5.5 ($\frac{5}{3}p - 6$, new proof)
Theorem 1 (Bazzi–Mitter)	Theorem 5.1 (no congruence condition)
Theorem 2 (1.77, $p \equiv 1 \bmod 4$)	Theorem 5.1 ($c < 1.7799$, all p)
Theorem 3 (1.62, $p \equiv 1 \bmod 4$)	Theorem 5.3 ($c < 1.62584$, all p)
Corollary 2 (1.62, all p)	Theorem 5.3
Conjecture 3 (self-dual / LCD)	Theorem 4.4 (proved)
Conjecture 4 (C')	Theorem 7.2 (proved)
Section 4: Theorem 4, Theorem 5, Lemmas 5–7	Remark 5.7 (removed)
Lemma 8 (Voloch)	Theorem 6.1 (with new case $p \equiv 5 \bmod 8$)
Theorem 6 (Voloch, Chebotarev)	Theorem 6.3
—	Lemma 3.4, Corollary 3.6, Proposition 4.10, Theorem 7.2 (new)
Conjecture (Bazzi–Mitter), restated	Conjecture 8.7 (refined, $p \equiv \pm 3 \bmod 8$)
—	Propositions 8.1, 8.3, Theorem 8.6 (new, 10 November 2026)

References

[BH17] N. Boston and J. Hao, The weight distribution of quasi-quadratic residue codes, arXiv:1705.06413.

[Ra98] E. M. Rains, Shadow bounds for self-dual codes, *IEEE Trans. Inform. Theory* 44 (1998), 134–139.